Appendix B

Explanatory Notes and Detailed Methods Report

1. Overview....34
   A. The Energy Information Administration's Quality Guidelines....34
   B. Concepts of Product Supply and Demand....34
2. Weekly Petroleum Supply Surveys....34
   A. Weekly Petroleum Supply Reporting System....34
   B. Weekly Supply Survey Methodology....35
      (1.) Sampling Frame....35
      (2.) Sample Design....35
      (3.) Collection....36
      (4.) Processing....36
      (5.) Imputation and Estimation....36
      (6.) Macro Editing....37
      (7.) Dissemination....37
   C. Additional Sources of Data....37
      (1.) Data Obtained Through Models....38
         a. Domestic Crude Oil Production (Tables 1 and 9)....38
         b. Exports (Tables 1, 7, 9)....38
         c. Stocks of Other Oils (Tables 1, 4, 9)....38
         d. Refinery Processing Gain (Tables 1, Line 20)....38
      (2.) Data Obtained from Supplemental Sources....39
         a. Natural Gas Liquids Production (Table 1, Line 16)....39
         b. Other Renewable Fuels/Oxygenate Plant Production (Table 1, Line 19)....39
         c. Other Supply Adjustment (Table 1, Line 25)....39
         d. Production of Finished Motor Gasoline Adjustment (Tables 2 and 9)....39
   D. Quality....39
      (1.) Response Rates....39
      (2.) Timing Issues....39
      (3.) Non-sampling Errors....39
      (4.) Resubmissions....39
      (5.) Revision Policy....40
   E. Petroleum Historic Stock Ranges....40
   F. Data Assessment....40
   G. Confidentiality—Data protection and disclosure - Weekly Supply Surveys....40
3. Weekly Petroleum Price Surveys....40
   A. Weekly Price Survey Methodology....40
      (1.) Sampling Frame....40
         a. EIA-878, "Motor Gasoline Price Survey"....40
         b. EIA-888, "On-Highway Diesel Fuel Price Survey"....41
      (2.) Sampling Design....41
         a. EIA-878, "Motor Gasoline Price Survey"....41
         b. EIA-888, "On-Highway Diesel Fuel Price Survey"....41
      (3.) Collection....42
      (4.) Processing and Micro Editing....42
      (5.) Imputation and Estimation....42
         a. EIA-878, "Motor Gasoline Price Survey"....42
         b. EIA-888, "On-Highway Diesel Fuel Price Survey"....42
      (6.) Macro Editing and Validation....43
         a. EIA-878, "Motor Gasoline Price Survey"....43
         b. EIA-888, "On-Highway Diesel Fuel Price Survey"....43
      (7.) Dissemination....43

B. Quality....43
(1.) Response Rates....43
(2.) Sampling and Non-sampling Errors....43
a. Sampling Errors....43
b. Non-sampling Errors....43
(3.) Revision Policy....44
C. Confidentiality—Data protection and disclosure for Weekly Price Surveys....44
4. Notes....44

Weekly Petroleum Status Report Explanatory Notes

Overview

The Energy Information Administration's Quality Guidelines

The data contained in the *Weekly Petroleum Status Report* (WPSR) are subject to separate information quality guidelines issued by the Office of Management and Budget (OMB), the Department of Energy (DOE), and Energy Information Administration (EIA). With available resources, EIA continually works to improve its systems in order to provide high quality information needed by public and private policymakers and decision makers. EIA has performance standards to ensure the quality (i.e., objectivity, utility, and integrity) of information it disseminates to the public. Quality is ensured and maximized at levels appropriate to the nature and timeliness of the disseminated information. Information about EIA's quality program is available at http://www.eia.doe.gov/smg/EIA-IQ-Guidelines.html.

Concepts of Product Supply and Demand

Petroleum supply estimates contained in the *WPSR* are often interpreted as an approximation of petroleum demand measured as product supplied. Product supplied is often called "implied" demand because it is a measure of demand that is implied by disappearance of petroleum products from facilities and activities in the "primary" supply chain. Facilities and activities in the primary supply chain include refineries and blending terminals, gas processing plants and fractionators, oxygenate producers, importers, exporters, bulk storage terminals, and pipelines. Total product supplied in the *WPSR* may be calculated from petroleum balances reported in Table 1. Total product supplied for crude oil and petroleum products is equal to crude oil input to refineries (line 14) + Other Supply Production (line 15) + net imports(line 21) - Stock Change (line 24) + Adjustment (line 25). Product supplied for individual products equals production plus imports minus stock change minus exports. "Crude Oil Supply Adjustment" (line 13) (formerly called "Unaccounted-for Crude Oil") is the balancing item between crude oil supply and disposition.

The secondary supply chain system is that portion of the overall distribution network that falls between producers and end-users. Product typically flows in bulk from the primary supply system into the secondary system before delivery in small quantities to consumers (the tertiary system). The secondary system includes storage at bulk plants; at retail motor fuel outlets, such as service stations, truck stops, and convenience stores; and at retail fuel oil dealers. Bulk plants are wholesale storage facilities that have less than 50,000 barrels of storage capacity and, by definition, receive product only by tank car or truck, not by barge, tanker, or pipeline. Tertiary inventories are held by end users and include fuel in vehicle tanks, heating oil in residential tanks, fuel oil held by utilities, jet fuel stored in facilities operated by end users, and certain proprietary storage of raw materials for the chemical industry (ethylene, propylene, etc.).

Data users sometimes consider demand as sales to the ultimate consumer or as the actual consumption of the product. Since there may be time delays between the movement of product into the primary market and its ultimate purchase or consumption, these definitions of demand require data on changes in secondary and/or tertiary stocks or the assumption that these values either remain constant or are small compared to primary supply. The most recent study of secondary stocks was done by the National Petroleum Council in 1989. This study revealed that secondary distillate stocks were equal to about 6.9 percent of distillate stocks and 6.7 percent of distillate storage capacity. The study also noted that secondary storage capacity was decreasing due to EPA regulations.

Weekly Petroleum Supply Surveys

The data presented in the *WPSR* include data collected by the EIA on seven weekly petroleum supply and two weekly petroleum price surveys and data released by Reuters Ltd. During the heating months (October through mid-March), data from a 3rd weekly price survey are included in Appendix D, "Winter Fuels Heating Prices."

Weekly Petroleum Supply Reporting System

The seven weekly petroleum supply surveys are part of the Petroleum Supply Reporting System (PSRS). The PSRS tracks the supply and disposition of crude oil, petroleum products, and natural gas liquids in the United States. The PSRS is organized into two data collection subsystems, the Weekly Petroleum Supply Reporting System (WPSRS) and the Monthly Petroleum Supply Reporting System (MPSRS). The WPSRS processes the data from the seven weekly surveys. The MPSRS includes eight monthly surveys and one annual survey. The survey forms that comprise the PSRS are:

1. EIA-800, "Weekly Refinery and Fractionator Report,"
2. EIA-801, "Weekly Bulk Terminal Report,"
3. EIA-802, "Weekly Product Pipeline Report,"
4. EIA-803, "Weekly Crude Oil Stocks Report,"
5. EIA-804, "Weekly Imports Report,"
6. EIA-805, "Weekly Bulk Terminal and Blender Report,"
7. EIA-809, "Weekly Oxygenate Report,"
8. EIA-810, "Monthly Refinery Report,"
9. EIA-812, "Monthly Product Pipeline Report,"
10. EIA-813, "Monthly Crude Oil Report,"
11. EIA-814, "Monthly Imports Report,"
12. EIA-815, "Monthly Bulk Terminal and Blender Report."
13. EIA-816, "Monthly Natural Gas Liquids Report"
14. EIA-817, "Monthly Tanker and Barge Movement Report"
15. EIA-819, "Monthly Oxygenate Report"
16. EIA-820, "Annual Refinery Report."

A copy of the forms and instructions is available at:
http://www.eia.gov/survey/

Weekly supply surveys are administered at seven key points along the petroleum production and supply chain: (1) refineries, fractionators, and gas processing plants, (2) bulk terminals, (3) product pipelines, (4) crude oil stock holders, (5) importers, (6) blenders and (7) fuel ethanol production facilities Monthly surveys also include inter- PAD District movements by pipelines, tankers, and barges. Weekly surveys do not capture petroleum movements. Data collected weekly using

Forms EIA-800 through EIA-805 and EIA-809 are similar to, though less detailed than, the data collected monthly using Forms EIA-810, EIA-812 through EIA-815 and EIA-819. Respondents reporting to the weekly surveys constitute a sample of those reporting on the monthly surveys.

Annual U.S. refinery capacity data are collected on the Form EIA- 820, "Annual Refinery Report." These data are published in the *Refinery Capacity Report*.

Weekly Supply Survey Methodology

Sampling Frame

The EIA weekly reporting system, as part of the Petroleum Supply Reporting System (PSRS), was designed to collect data similar to those collected monthly. The sample of companies that report weekly in the WPSRS are selected from the universe of companies that report on the corresponding monthly forms with the exception of the EIA-801 in 2010.

The sampling frame for Form EIA-800 "Weekly Refinery Report" includes refineries reporting on Form EIA-810 "Monthly Refinery Report" as well as fractionators reporting on Form EIA-816 "Monthly Natural Gas Liquids Report." Monthly reports on Form EIA-810 are required from operators of every operating and idle refinery located in the 50 States, District of Columbia, Virgin Islands, Puerto Rico, and other U.S. territories. Monthly reports on Form EIA-816 are required from operators of every operating and idle gas processing plant, fractionator, and butane isomerization plant located in the 50 States and the District of Columbia.

The EIA-801 sampling frame consists of all companies reporting ending stocks on the EIA-815, "Monthly Bulk Terminal and Blender Report." This includes every bulk terminal and blending facility operating company located in the 50 States, the District of Columbia, Puerto Rico, and the Virgin Islands. A bulk terminal is primarily used for storage and/or marketing of petroleum products and has a total bulk storage capacity of 50,000 barrels or more, and/or receives petroleum products by tanker, barge, or pipeline. Bulk terminal facilities associated with a product pipeline are included.

The EIA-802 sampling frame consists of all companies reporting on the EIA-812, "Monthly Product Pipeline Report." This includes all petroleum product pipeline companies that transport refined petroleum products (including interstate, intrastate, and intracompany pipeline movements) in the 50 States and the District of Columbia. Bulk terminal facilities associated with a product pipeline are excluded.

The EIA-803 sampling frame consists of all companies reporting on the EIA-813, "Monthly Crude Oil Report." This includes all companies that carry or store 1,000 barrels or more of crude oil. Included are gathering and trunk pipeline companies (including interstate, intrastate, and intracompany pipelines), crude oil producers, terminal operators, storers of crude oil (except refineries), and companies transporting Alaskan crude oil by water in the 50 States and the District of Columbia.

The EIA-804 sampling frame consists of all companies reporting on the EIA-814, "Monthly Imports Report." This includes each Importer of Record (or Ultimate Consignee in some situations regarding Canadian imports) that import crude oil or petroleum products (1) into the 50 States and the District of Columbia, (2) into Puerto Rico, the Virgin Islands, Guam and other U.S. possessions (Midway Islands, Wake Island, American Samoa, and Northern Mariana Islands), (3) Foreign Trade Zones located in the 50 States and the District of Columbia and (4) from Puerto Rico, the Virgin Islands and other U.S. possessions into the 50 States and the District of Columbia.

The EIA-805 sampling frame consists of all companies reporting inputs and production on the EIA-815, "Monthly Bulk Terminal and Blender Report." This includes all storage terminals which produce finished motor gasoline through the blending of various motor gasoline blending components, natural gas liquids, and oxygenates in the 50 States, the District of Columbia, Puerto Rico, the Virgin Islands, Guam and other U.S. possessions.

The EIA-809 sampling frame consists of all operators of facilities reporting fuel ethanol production on the EIA-819, "Monthly Oxygenate Report." This includes fuel ethanol production facilities in the 50 States and the District of Columbia.

Sample Design

The sampling procedure used for the weekly surveys is the cut- off method. In the cut-off method, companies are ranked from largest to smallest on the basis of quantities reported during some previous period. Companies are chosen for the sample beginning with the largest and adding companies until the total sample covers approximately 90 percent of the total volumes for each item and each geographic region for which data may be published. For example, for distillate fuel oil stocks, the weekly sample includes those respondents whose combined volumes of stocks for distillate fuel oil from refineries, bulk terminals, and pipelines constitute at least 90 percent of the total volume of distillate fuel oil stocks as reported in the corresponding monthly surveys.

To assure 90-percent coverage of the total for each item collected and each geographic region for each weekly survey, the sample is reviewed each month. This review focuses on changes in the current monthly data as it relates to the weekly surveys, changes in the weekly surveys that impact the monthly surveys, and changes in respondent reporting patterns. Companies are added or removed from the surveys based on the changes. Refer to Table B1 for sample size of weekly surveys.

For the weekly surveys, better coverage will most likely reduce any sampling error. As shown in Table B2, 2012 coverage was comparable to 2011. Of the 21 product and supply type combinations, all except one had coverage above 90 percent in 2012. For 3 of the 21 combinations, 2012 coverage increased slightly over 2011. The largest percentage increase from 2011 to 2012 was for jet fuel oil imports, an increase of 0.4 percent. Bulk Terminals distillate fuel oil stocks and residual fuel oil production had the largest percentage decrease from 2011 to 2012, a decline of 6 percent each. Tabulations were done before rounding of the coverage values. Total motor gasoline production percentages include production from refineries, terminals and blenders.

Collection

Table B1. Frame and Sample Size for Weekly Supply Surveys

	Weekly Form	July 2026 Frame Size	Weekly Sample Size
Refiners (Refineries)	EIA-800	132	111
Bulk Terminals	EIA-801	*	*
Product Pipelines	EIA-802	108	59
Crude Oil Stock Holders	EIA-803	229	92
Importers	EIA-804	275	117
Terminal Blenders	EIA-805	1,461	868
Natural Gas Liquids Report	EIA-806	443	189
Ethanol Producers	EIA-809	253	145

* The sample for the EIA-801 comes from the same frame as the EIA-805.

Table B2. Average Coverage for Weekly Surveys, 2011 and 2012
(Percent of Final Monthly Volumes Included in Monthly-From-Weekly Sample)

Product	Stocks (%)						Production (%)		Imports (%)	
	Refinery		Bulk Terminals		Pipeline		2012	2011	2012	2011
	2012	2011	2012	2011	2012	2011				
Total Motor Gasoline	99	100	89	93	98	99	98	99	97	97
Jet Fuel	99	100	92	94	99	100	100	100	100	99
Distillate Fuel Oil	99	100	86	91	98	98	98	99	96	96
Residual Fuel Oil	99	100	89	93	-	-	99	100	96	99
Crude Oil	99	99	-	-	-	-	-	-	98	99

Survey data for the WPSR are collected by facsimile, Internet using secure file transfer, and electronic transmission on a weekly basis. All respondents must submit their data by 5:00 p.m. on the Monday following the end of the report period. The weekly report period begins at 7:01 a.m. on Friday and ends at 7:00 a.m. on the following Friday.

Processing

Data collected through the WPSRS are received, logged into an automated Survey Control File, keyed, and processed through an edit program. Cell values determined to be unusual or inconsistent with other cell values are flagged either by automated process or analyst review. The validity of the value of each flagged cell is investigated. From the investigation, some flagged values are either verified or corrected by the respondent. Any remaining flagged values are referred to as unresolved. Imputation is performed for nonrespondents and unresolved data items. The cleansed data are further reviewed at the aggregate level to determine if other data issues exist (see Macro Editing).

A clean data file is available by the close of business Tuesday. Corrections to previous periods, late submissions, or resubmissions for the current period received after publication are used in editing and imputation for the following periods (see Revision Policy).

Imputation and Estimation

After company reports have been checked and entered into the weekly database, values are imputed for companies that have not responded, reported incomplete data, or reported data that failed editing and could not be confirmed. The imputed values are calculated using exponentially smoothed means of recent weekly reported values for this specific company.

The equation for the exponential smoothing is:

$$Y_t = \alpha * y_t + (1 - \alpha) * Y_{t-1}$$

where

Y_t is the prediction for week t+1 (using data through week t),

y_t is week t's reported value,

Y_{t-1} is the prediction for week t (using data through week t-1),

α is a number between 0 and 1, chosen by survey/product/type

In the equation for exponential smoothing, the size of α controls the importance of last week's value relative to the aggregate of all weeks before that as represented by the prediction for last week. For example, if $\alpha = 0.8$, then last week's value is much more important in predicting this week's value than all the previous week's values are since the weight of last week is 0.8 and the weight of the previous weeks collectively is 0.2. In general, the α values for the expected means of the non-zero

responses are low for imports (last week is much less important than history) and much higher for production, inputs and stocks.

The imputed values are treated like reported values in the estimation procedure, which calculates ratio estimates of the weekly totals. First, the current week's data for a given product reported by companies in a geographic region are summed (weekly sum, W_s.) Next, the most recent month's data for the product reported by those same companies are summed (monthly sum, M_s.) Finally, the most recent month's data for the product as reported by all companies, including adjustments made in the monthly process, is summed (M_t). The current week's ratio estimate for that product for all companies, W_t, is given by:

$$W_t = (M_t / M_s) * W_s$$

The ratio (M_t / M_s) may be adjusted to account for very unusual events or industry changes not yet reflected in the lagged monthly data. For example, the hurricanes in September 2005 rendered the September data unrepresentative for purposes of applying the ratio to the *WPSR* in December 2005. Note, however, the gasoline and ethanol fuel adjustment is not included in M_t and is treated explicitly.

This procedure is used directly to estimate total weekly inputs to refineries and production. When refineries are closed or inoperable, the lagged monthly data impacts the estimate of operable capacity and percent utilization in the WPSR. Operable capacity is the latest reported monthly operable capacity. The percent utilization is calculated as gross weekly inputs divided by operable capacity. The use of monthly capacity data may result in an overestimate of operable capacity and an underestimate of percent utilization until the shutdown is shown in the monthly data.

To estimate stocks of finished products, the preceding procedure is followed separately for refineries, bulk terminals, and pipelines. Total estimates are performed by summing over establishment types.

Published values of gasoline production include a fuel adjustment to account for the imbalance between supply and disposition of motor gasoline blending components and fuel ethanol. For further detail, refer to Additional Sources of Data, Data Obtained from Supplemental Sources (below).

Weekly imports data are highly variable on a company-by-company basis or a week-to-week basis. Therefore, an exponentially smoothed ratio has been developed for weekly imports. The estimate of total weekly imports is the product of the smoothed ratio and the sum of the weekly reported values and imputed values.

For imports, the ratio is smoothed as follows:

$$R_t = \alpha * r_t + (1 - \alpha) * R_{t-1}$$

where

R_t is the smoothed ratio for week t+1 (using ratios through week t),

r_t is week t's ratio of the most recent monthly total for all respondents to the monthly total of respondents from the weekly sample,

R_{t-1} is the smoothed ratio for week t (using ratios through week t-1),

α is a number between 0 and 1, chosen by product but not by PADD/Respondent ID.

When $M_s = 0$, then r_t is not defined for the week and the smoothed ratio is not updated, that is, the previous smoothed ratio is used as the multiplier.

Macro Editing

After the respondent-level data have been collected and processed. The WPSR processing system is "locked down" to all staff except a select group of industry analysts and statisticians, referred to as the WPSR Review Team. Aggregate-level estimates are generated by product and geographic region for the current week, three prior weeks, year ago data for the same week, along with 4-week averages. The WPSR Review Team has the responsibility for reviewing the aggregated data for all products and resolving inconsistencies with these estimates.

Once the WPSR Review Team have completed their review, preliminary WPSR tables are generated and provided to the Petroleum Division Director (PDD) for review. At 4 p.m., the team meets with the PDD for a final review and discussion of the estimates. Discrepancies in the data are discussed and, if necessary, adjustments are made and the final published statistics are generated for release on Wednesday morning at 10:30 a.m.

Dissemination

The data are published in the *WPSR* and the *TWIP* every Wednesday for the report period ending on the previous Friday. The WPSR tables are released to the EIA Web site at 10:30 a.m. (Eastern Standard Time) in CSV and XLS formats. The weekly highlights are released in PDF format at 10:30 am. The entire *WPSR* is released at 1:00 p.m. in PDF and HTML format. For weeks which include holidays (or have other disruptions to normal operations), releases are delayed by one day. The *WPSR* tables can be accessed at: http://www.eia.gov/petroleum/supply/weekly/.

Selected data from the weekly supply surveys are also published in the *This Week in Petroleum (TWIP)* generally available at 1:00 p.m. Eastern Time on Wednesdays. The *TWIP* can be accessed at: http://www.eia.gov/oog/info/twip/twip.asp.

Additional Sources of Data

Due to the tight time constraints in publishing weekly petroleum supply statistics and the desire to reduce industry response burden, some of the statistics published in the *WPSR* are obtained from sources other than the 7 weekly supply surveys. These other sources include models to data and data from supplemental sources such as the *PSM* or the Bureau of the Census.

Data Obtained Through Models

Domestic Crude Oil Production (Tables 1 and 9)

EIA estimates weekly domestic crude oil production using a combination of short-term forecasts and the latest available production estimates from Alaska. The four data elements contributing to the estimate are:

- the most recent Short-Term Energy Outlook (STEO) model estimate (including interim estimates) for average daily production for the lower 48 States and the Federal Gulf of America (GOA) (STEO Table 4a: http://www.eia.gov/forecasts/steo/data.cfm?type=tables);
- daily production volumes delivered from the North Slope of Alaska to the Trans-Alaska Pipeline System (TAPS) (reported to EIA by the Alyeska Pipeline Service Company);
- daily volumes of natural gas plant liquids produced on the North Slope delivered to TAPS (reported to EIA by BP); and
- daily production for South Alaska estimated from monthly production reports (lagged by two months) from the Alaska Oil and Gas Conservation Commission (AOGCC).

Most of the uncertainty in the weekly estimate is associated with the STEO forecast for lower 48 and GOA production. For example, when lower 48 crude oil production is either increasing or decreasing rapidly, the accuracy of the estimate for any particular month is likely to be reduced. During tropical storms or hurricanes that affect Gulf of America oil production, near real-time daily shut-in volumes reported by the Bureau of Safety and Environmental Enforcement (BSEE) are subtracted from the daily lower 48 estimate. See Previous STEO Forecasts at http://www.eia.gov/forecasts/steo/data.cfm?type=tables for comparisons of current and previous monthly STEO forecasts for lower 48 oil production and percent changes for STEO quarterly forecasts.

The weekly estimates of domestic crude oil production are reviewed monthly when the *Short-Term Energy Outlook* (STEO) is released to identify differences with recent trends in survey-based domestic production reported in the *Petroleum Supply Monthly* (PSM) and other current data. If a large difference between the two series is observed, the weekly production estimate may be re-benchmarked on weeks when the STEO is released.

Exports (Tables 1, 7, and 9)

Official U.S. exports statistics for crude oil, petroleum products, and biofuels are compiled and published by the U.S. Census Bureau each month. EIA obtains these data on a monthly basis approximately six weeks after the end of the monthly reporting period. Weekly exports statistics for crude oil, petroleum products, and biofuels are obtained through a memorandum of understanding with the U.S. Customs and Border Protection (CBP) to access their Automated Commercial Environment (ACE) system. Except for exports to Canada, all entities exporting products from the United States are required to file export information with ACE. EIA receives unedited data files from CBP each week that EIA processes to remove duplicate entries and to convert measurements as needed. This edited data is then used to estimate U.S. weekly exports of crude oil, petroleum products, and biofuels. For exports to Canada, EIA also applies an Unobserved Components Model as part of the weekly estimate where applicable. For residual fuel oil and other oils, EIA further applies a post processing regression method. The weekly exports estimates are updated each week given the availability of current CBP data. In weeks when CBP data is not available, the most recent four-week average of exports will be used. For more detail, please see Appendix D, the "Statistical Methodology of Estimating Petroleum Exports Using Data from U.S. Customs and Border Protection."

Stocks of Other Oils (Tables 1, 4, 9)

Stocks of "other oils" are derived from a combination of both weekly survey data and estimates based on prior monthly reported values. Other oils stocks include weekly survey data collected for natural gas plant liquids (NGPLs) and liquid refinery gases (LRGs) excluding propane/propylene which is reported separately, unfinished oils, kerosene, and asphalt/road oil. These products typically account for the majority of other oils stocks. Stocks of the remaining minor products included in other oils inventories not collected on weekly survey forms are estimated. Minor products include aviation gasoline, other hydrocarbons and oxygenates, aviation gasoline blending components, naphtha and other oils for petrochemical feedstock use, special naphtha, lube oils, waxes, coke, and miscellaneous oils. An estimate of weekly stocks of minor products is derived by first computing an average daily rate of stock change for the minor products for each month based on monthly data for the past 6 years (Table 1 of the *PSM*). The daily stock change for a month is estimated by subtracting the prior month's end of month other oils stocks from the current month's end of month other oils stocks and dividing by the number of days in the current month. This average daily rate and the minor stock levels from the most recent *PSM* are then used to estimate the minor product stock level for the current week.

Since some of the components of the stocks of other oils are based on values from past monthly data, analysts review the estimate to determine if factors such as recent increases or decreases in crude runs or reported outlier data require an adjustment to the estimate of stocks of minor products.

Refinery Processing Gain (Table 1, Line 20)

Processing gain is the volumetric amount by which total output is greater than input for a given period of time. This difference is due to the processing of crude oil into products which, in total, have a lower specific gravity than the crude oil processed.

Processing gain in the *WPSR* is calculated by dividing processing gain from Table 29 of the *PSM* by Refinery and Blender Net Inputs of Crude Oil in thousands of barrels per day from Table 3 of *PSM* for each of the latest 12 months of the *PSM*. The 12 values are added and divided by 12. The result is then multiplied by this week's crude oil input to refineries value in Table 1 of the *WPSR* to obtain the processing gain value for the week.

Data Obtained from Supplemental Sources

Natural Gas Plant Liquids Production (Table 1, Line 16)

Natural Gas Plant Liquids Production is not collected on the weekly surveys. The volume shown for "Natural Gas Plant Liquids Production" is "Field Production" of "Natural Gas Plant Liquids and Liquefied Refinery Gases" from Table 3, "U.S. Daily Average Supply and Disposition of Crude Oil and Petroleum Products" of the latest *PSM*. For further information see the Explanatory Notes in the appendix of the *PSM* available at: http://www.eia.gov/petroleum/supply/monthly/

Other Renewable Fuels/Oxygenate Plant Production (Table 1, Line 19)

"Other Renewable Fuels/Oxygenate Plant Production" is derived from data on Table 3 of the latest PSM. It is derived by adding Total "Renewable Fuels and Oxygenate Plant Net Production," less Renewable Fuels and Oxygenate Plant "Fuel Ethanol" production, plus the adjustments to "Oxygenates (excluding fuel ethanol) and adjustments to "Renewable Fuels Except Fuel Ethanol." Other Renewable Fuels/Oxygenate Plant Production includes production of "Oxygenates (excluding fuel ethanol)" and "Renewable Fuels Except Fuel Ethanol." "Oxygenates (excluding fuel ethanol)" include ETBE, MTBE, and E85 as well as input of denaturants for fuel ethanol at fuel ethanol plants. For further information see the explanatory notes in the appendix of the *PSM* available at: http://www.eia.gov/petroleum/supply/monthly/

Other Supply Adjustment (Table 1, Line 25)

Other Supply Adjustment is equal to the sum of the "Adjustment" for Refiner and Blender Net Production of Finished Motor Gasoline from Table 2 of the *WPSR* and the adjustments to the supply for "Hydrogen" and "Other Hydrocarbons" from Table 3 of the *PSM*.

Production of Finished Motor Gasoline Adjustment (Tables 2 and 9)

Production of finished motor gasoline reported in Tables 2 and 9 of the *WPSR* includes refinery production, blender production, and adjustments to account for imbalances between supply and disposition of motor gasoline blending components and fuel ethanol. An adjustment is needed to finished motor gasoline production because there typically is more supply than disposition reported for motor gasoline blending components and fuel ethanol. Since there is no end-user demand for motor gasoline blending components or fuel ethanol, the imbalance is typically interpreted as unreported gasoline production at blenders. Gasoline production adjustments are included in Total US finished gasoline production reported in the *WPSR*. The adjustment is the sum of the values required to balance the supply and disposition of motor gasoline blending components and fuel ethanol. Supply is production plus imports minus stock change. Disposition is refinery and blender net production plus exports.

For motor gasoline blending components, production equals the motor gasoline blending component adjustment value from Table 3 of the *PSM*. Imports, stock change, and refinery and blender net inputs are current weekly data; and exports are from the Petroleum Export Model. For fuel ethanol, production equals ethanol plant production, imports, stock change, and refinery and blender net inputs are current weekly data and exports are from the Petroleum Export Model.

Additional details concerning gasoline adjustments are available in Appendix B, "Detailed Statistics Explanatory Notes" of the *PSM*.

Quality

Response Rates

The response rate for the weekly supply surveys is generally 95 to 100 percent. Chronic nonrespondents and late filing respondents are contacted by telephone and reminded of their requirement to report. Nearly all of the major companies report on time. The nonresponse rate for the published estimate is usually between 1 percent and 2 percent.

Timing Issues

Timing of reported data can impact published results. For example, the calculation of product supplied includes imports and change in stock levels. Normally imports would result in a stock increase. However, respondents recording inventories are frequently different than the respondents reporting imports. The accounting system of one respondent may lag that of another, resulting in the imports and associated stocks being reported in different weeks. These timing differences result in weekly variations in product supplied.

Non-sampling Errors

The weekly supply data are closely watched by market analysts and are sometimes attributed to movements in both spot and futures prices on the day the data are released. When petroleum markets are particularly tight or when the data are not what the market is expecting, (e.g. a build in inventories occurs when a decline is expected), the weekly data take on a more significant role in the assessment of petroleum markets, where such assessments affect billions of dollars in the financial markets.

Non-sampling errors may arise in the survey estimates from a number of sources including: (1) the inability to obtain data from all companies in the frame or sample (non-response and the method used to account for non-response), (2) response errors, (3) differences in the interpretation of questions or definitions, (4) mistakes in recording or coding of the data obtained from respondents, (5) data timing, and (6) other errors of collection, response, coverage, and estimation.

Resubmissions

Resubmissions are required whenever an error greater than 5 percent of the true value is discovered or if requested by EIA. Late submissions or resubmissions received after the publication date are used for editing and imputation for future periods. In rare instances, the data are used to publish a revised estimate. See Revision Policy below.

Revision Policy

EIA will disseminate revised weekly data only if the revision is expected to substantively affect understanding of U.S. petroleum supplies. The decision to disseminate a revision to weekly data will be based on EIA's judgment of the revision's expected effect. If a revision is necessary, it will be disseminated in the next regularly scheduled release of the weekly products.

Petroleum Historic Stock Ranges

The 5-year high/low stock ranges displayed in Figures 1 through 6 are provided to help WPSR users compare current petroleum inventories to recent historic levels on a U.S. total and regional basis.

The 5-year ranges provide the reader with the highest and lowest weekly stock levels for a given product by region over the equivalent week during the prior five years. Current weekly stock estimates published in the WPSR (labeled Weekly) are plotted in relation to these 5-year stock levels (shaded area on the charts) for crude oil, total motor gasoline, distillate fuel oil, kerosene-type jet fuel, residual fuel oil, and propane inventories. The charts show two years of data, covering periods either from December through December or June to June.

Data Assessment

The principal objective of the PSRS is to provide an accurate picture of petroleum industry activities and of the availability of petroleum products nationwide from primary distribution channels. The weekly data, which are based on sample estimates stemming largely from preliminary company data, serve as leading indicators of the monthly data. The weekly data are not expected to have the same level of accuracy as the preliminary monthly data when compared with final monthly data. However, the weekly data are expected to exhibit like trends and product flow characteristic of the preliminary and final monthly data.

To assess the accuracy of weekly statistics, monthly estimates derived from weekly estimates are compared with the final monthly aggregates published in the Petroleum Supply Annual (PSA). Although final monthly data published in the PSA are still subject to error, they have been thoroughly reviewed and edited, they reflect all revisions made during the year, and they are considered to be the most accurate data available. The mean absolute percent error provides a measure of the average revisions relative to the aggregates being measured for a variable. The mean absolute percent error for 2007 weekly data was less than 2 percent for 22 of the 62 major petroleum variables analyzed. As a group, stocks continued to have the most accurate monthly from weekly estimates. The detailed analysis is available in a feature article entitled "Accuracy of Petroleum Supply Data" available at: http://www.eia.doe.gov/petroleum/supply/monthly/ archive/2009/2009_02/pdf/art0902.pdf

Confidentiality—Data protection and disclosure - Weekly Supply Surveys

The information reported on Forms EIA-800 through EIA-805 and EIA-809 is kept confidential and not disclosed to the public to the extent that it satisfies the criteria for exemption under the Freedom of Information Act (FOIA), 5 U.S.C. §552, the DOE regulations, 10 C.F.R. §1004.11, implementing the FOIA, and the Trade Secrets Act, 18 U.S.C §1905. The Energy Information Administration (EIA) protects this information in accordance with its confidentiality and security policies and procedures.

The Federal Energy Administration Act requires the EIA to provide company-specific data to other Federal agencies when requested for official use. The information reported on these forms may also be made available, upon request, to another component of the Department of Energy (DOE); to any Committee of Congress, the General Accounting Office, or other Federal agencies authorized by law to receive such information. A court of competent jurisdiction may obtain this information in response to an order. The information may be used for any nonstatistical purposes such as administrative, regulatory, law enforcement, or adjudicatory purposes.

Disclosure limitation procedures are not applied to the statistical data published from these surveys' information. Thus, there may be some statistics from forms EIA-800 through EIA-805 and EIA-809 that are based on data from fewer than three respondents, or that are dominated by data from one or two large respondents. In these cases, it may be possible for a knowledgeable person to estimate the information reported by a specific respondent.

Company specific data are also provided to other DOE offices for the purpose of examining specific petroleum operations in the context of emergency response planning and actual emergencies.

Weekly Petroleum Price Surveys

Weekly Price Survey Methodology

EIA survey price data contained in this report are derived from two weekly surveys, the EIA-878, "Motor Gasoline Price Survey," and the EIA-888, "On-Highway Diesel Fuel Price Survey." These surveys provide timely information on national and regional retail prices of gasoline and on-highway diesel fuel.

Sampling Frame

EIA-878, "Motor Gasoline Price Survey"

The target population is all active retail gasoline outlets in the United States for a given week. The population includes two types of outlets—big-box and non-big-box outlets. Big-box outlets typically sell large volumes of gasoline at discounted prices.

The EIA-878 sample was drawn from a frame of approximately 130,000 retail gasoline outlets in the United States that were active in 2016. The gasoline outlet frame was constructed by combining outlet information from a private commercial source with information contained on existing EIA petroleum product frames and surveys, federal and state administrative records, and other publicly available sources. Outlet names, physical addresses, and ZIP codes were

obtained from the private commercial data source. The individual outlets in the frame were assigned to counties after converting the physical addresses to geographic coordinates. Each outlet is designated as either in an area requiring reformulated gasoline (RFG) based on Environmental Protection Agency (EPA) program requirements or in an area designated as a conventional gasoline area. Reformulated gasoline is required by EPA in any area that is designated as an ozone nonattainment area. A conventional area is defined as any area that does not require the sale of reformulated gasoline. All formulations of finished motor gasoline may be sold in conventional areas. The outlets were then further assigned to city areas based on the geographic areas as defined by EIA.

EIA-888, "On-Highway Diesel Fuel Price Survey"

The target population is all active retail on-highway diesel fuel outlets in the contiguous United States for a given week. Due to statistical and operational considerations, outlets in the States of Alaska and Hawaii were excluded from the target population. The population includes two types of outlets—truck stops and service stations that sell on-highway diesel fuel. For the sole purpose of sampling efficiency, we define a truck stop as an on-highway diesel fuel retail outlet that has diesel fuel bays designed to accommodate and serve large trucks and may also offer amenities such as restaurants, showers, truck maintenance and repair, and laundry services. Truck stops typically sell larger volumes of diesel fuel and have designated diesel fueling bays that are separate from gasoline or diesel fuel pumps intended for automobiles and light trucks. In contrast, service stations sell diesel fuel in the same area as gasoline.

The EIA-888 sample was drawn from a frame of approximately 73,000 service stations and 9,500 truck stops in the contiguous United States that were active in 2021. We constructed the diesel fuel outlet frame by combining information from a private commercial source with information contained on our existing petroleum product frames and surveys and other publicly available sources. Auxiliary data at the outlet level on the frame were used to stratify truck stops on the frame and included truck diesel fuel lane counts, traffic volumes on nearby roadways, truck parking availability at the outlet, and sales of diesel exhaust fluid (DEF) from a pump.

Sample Design

EIA-878, "Motor Gasoline Price Survey"

The gasoline outlet sample implemented on May 14, 2018, is a stratified systematic sample with a total size of 1,000 retail outlets. Retail gasoline outlets are assigned to primary sampling strata based on physical address. These primary sampling strata are non-overlapping, and one or more primary sampling strata may be combined to correspond to a publication cell. A publication cell is defined by geography (PADD, state, and city) and attainment status (reformulated or conventional gasoline). Hence, New York State reformulated gasoline is a publication cell. New York City, conventional gasoline in PADD 1A (New England), and all of the United States are also publication cells.

The primary sampling strata are further substratified by retail gasoline outlet type (big-box or non-big-box). The total sample size is allocated to the sampling substrata in proportion to the number of outlets in the cell after weighting the big-box substrata in recognition of larger annual sales volume per outlet compared with non-big-box substrata. Sampling within each sampling substratum is performed by ordering the outlets by county and ZIP code and selecting an independent systematic random sample without replacement. This procedure results in adequate sample representation by ZIP code within a given substratum.

Based on annual assessments of the gasoline frame, the sample may be augmented to account for new outlets that are established. In addition, some geographic regions may experience relatively higher annual rates of outlets going out of business. Those geographic regions with relatively higher rates of sample attrition may be oversampled for newly identified outlets, compared with other regions, to help offset these smaller sample sizes.

EIA-888, "On-Highway Diesel Fuel Price Survey"

The on-highway diesel fuel sample implemented on June 13, 2022, is a stratified systematic sample with a total size of 590 retail outlets. Retail diesel fuel outlets are assigned to eight primary sampling strata based on physical address. These primary sampling strata are nonoverlapping, and they correspond to the published regional breakouts of Petroleum Administration for Defense Districts (PADDs), the most detailed geographic levels used to define publication cells and include PADDs 2, 3, 4, three sub-PADDs within PADD 1, and the two subparts of PADD 5 (the State of California and the West Coast region excluding California). The U.S., the East Coast (PADD 1), and the West Coast (PADD 5) are considered secondary publication cells because they include primary publication cell components. Each primary sampling stratum is further substratified into a certainty substratum of truck stops and up to four noncertainty substrata: large truck stops, medium truck stops, other truck stops, and service stations. Truck stops that are selected in certainty substrata are included in the sample with a probability of 1. By substratifying the primary strata, 38 sampling strata were formed.

We developed a model using historical sales data collected with Form EIA-821, "Annual Fuel Oil and Kerosene Sales Report," to estimate annual diesel fuel sales volumes at the outlet level. This model helped us stratify the truck stops into up to four substrata for a given primary sampling stratum based on auxiliary data at the outlet level on the frame.

We allocated sample sizes to the noncertainty substrata based on the substratas' relative estimated diesel fuel sales volumes, subject to constraints on minimum sample sizes and maximum sampling weights. Sampling within each noncertainty sampling substratum is performed by ordering the outlets by county and ZIP code and selecting an independent systematic random sample without replacement using a fractional interval. This procedure results in adequate sample representation by ZIP code within a given substratum.

Based on annual assessments of the diesel fuel frame, the sample may be augmented to account for new outlets that we identified since the construction of the initial frame. In addition, some geographic regions may experience relatively higher annual rates of outlets going out of business. Those geographic regions with relatively higher rates of sample attrition may be oversampled for newly identified outlets, compared with other regions, to help offset these smaller sample sizes.

Collection

Each Monday, the individual gasoline and diesel outlets are called and asked to report the pump price of their products including taxes as of 8:00 a.m. local time. If Monday is a holiday, the calls are made on the next business day; however, the Monday price is still recorded. The collection takes place using a computer assisted telephone interview (CATI) with built-in editing. Companies who prefer to report through their headquarters on behalf of their selected outlets are allowed to do so. We also collect weekly prices via email, text, fax, web survey, or manual retrieval from company websites. Data obtained through non-phone methods are entered into the CATI system and treated the same as phone-collected prices. Nonrespondent firms are telephoned several times. The data may be collected more frequently during emergency situations.

For the outlets selected in the gasoline and diesel samples, we collect their most recently available annual sales volume data only once when these outlets are selected for a new sample. The data are collected by mail or electronic transmissions. We use the volume data only to calculate our published volume-weighted average diesel fuel price estimates, and we do not publish these data.

For the few outlets in the sample that report selling on-highway diesel fuel to automobiles and trucks at different prices, we collect only the weekly truck prices from these outlets and weight these prices by the recent annual truck sales volumes collected from Form EIA-888, "On-Highway Diesel Fuel Price Survey Schedule B." We do not collect the weekly automobile diesel prices for these outlets because we do not believe that these automobile sales would have a meaningful impact on the published price estimates to justify the additional response burden.

In 2007, on-highway diesel fuel prices were collected for two types of diesel fuel, ultra-low sulfur and low sulfur. This dual collection was in response to the industry's implementation of EPA requirements phasing out the use of low sulfur diesel fuel. Publication of Low Sulfur On-Highway Diesel (LSD) prices at the U.S. level was discontinued on December 8, 2008 due to a diminishing number of stations selling LSD as a result of EPA diesel fuel regulations. EIA continued to collect LSD prices from retail outlets and included them in the Diesel Average All Types price until July 26, 2010, when no more outlets reported LSD sales. Beginning July 26, 2010 publication of the Ultra-Low Sulfur Diesel (ULSD) price became fully represented by the Diesel Average All Types price. As of December 1, 2010 (September 1, 2006 in California), any on-highway diesel fuel sold is ULSD as mandated by EPA on-highway diesel fuel regulations.

Processing and Micro Editing

The data are edited when they are entered into the CATI system, normally during the phone interview. Respondents are asked to verify prices that fail edits. If prices are outside a certain range or fail other criteria (e.g., the price of a station's fuel grade is the same or less than the price of a lower grade), respondents are also asked to explain the reason for the extreme deviation in price. Data obtained through non-phone methods are also entered into the CATI system. If the data fail the edits, the respondents are called and asked to verify their reported price(s). Imputation is used for outliers and nonrespondents.

In addition, in the middle of the weekly data collection, quality checks are performed on collected data. This is done to test the integrity of the current data, check for severe fuel price changes, and re-set any records which have been resolved. Outliers discovered during the pre-check are re-called to recheck or correct these prices. Any edits introduced to the data by this process will be applied when another pre-check or final processing is run.

Final processing takes place once all records in the CATI system have been resolved. Many of the same tasks of the pre-check process are repeated and final price estimates are created.

Imputation and Estimation

EIA-878, "Motor Gasoline Price Survey"

We handle item and unit nonresponse to weekly gasoline prices and annual sales volumes at the outlet level by imputation. Depending on available information, the imputation procedure is based on a model that incorporates some combination of previous survey data reported by the outlet, survey data reported by similar outlets in the sample, and data obtained from a private commercial source.

The estimation for weekly prices uses two sources of data from the Motor Gasoline Price Survey: annual sales volumes for each outlet in the sample and weekly price data for those outlets. Prior to implementing the new weekly sample, EIA collected annual sales volumes and ethanol content for regular, midgrade, and premium gasoline for the retail gasoline outlets in the sample from owners of the outlets and top suppliers of retail gasoline.

The sampling weight for a given sampled outlet is the reciprocal of the outlet's probability of selection in the sample. Using the annual sales volume data to estimate average prices, the volume weight for a given sampled outlet was constructed by multiplying its sampling weight by its annual sales volume. These volume weights are applied each week to the reported or imputed outlet gasoline prices to obtain weighted average price estimates for the formulations, grades, and geographic areas that EIA publishes. For quality assurance purposes, average price estimates are withheld from publication if at least half of the weighted annual sales volume is based on outlets for which the weekly gasoline prices are imputed.

EIA-888, "On-Highway Diesel Fuel Price Survey"

We handle item and unit nonresponse to weekly diesel fuel prices and annual sales volumes at the outlet level by imputation. Depending on available information, the imputation procedure is based on a model that incorporates some combination of previous survey data reported by the outlet, survey data reported by similar outlets in the sample, and data obtained from a private commercial source.

The estimation for weekly prices uses weekly price data for outlets in the sample that are collected from Form EIA-888, "On-Highway Diesel Fuel Price Survey Schedule A," and recent annual sales volumes for each of these outlets that are collected from Form EIA-888, "On-

Highway Diesel Fuel Price Survey Schedule B." The sampling weight for a given sampled outlet is the reciprocal of the outlet's probability of selection in the sample. To estimate average prices, we first calculate the weighted volume for a given sampled outlet by multiplying its sampling weight by its annual sales volume. We then apply these volume weights each week to the reported or imputed outlet diesel fuel prices to obtain weighted average price estimates for the geographic areas that we publish. For quality assurance purposes, we withhold average price estimates from publication if at least half of the weighted annual sales volume is based on outlets for which the weekly diesel fuel prices are imputed.

Macro Editing and Validation

EIA-878, "Motor Gasoline Price Survey"

Once the motor gasoline price data have been processed, the data are checked through a validation program. The program identifies the outliers in price changes from a week ago and in actual prices by grade and region. Significant outliers are investigated and verified by calling the respondent(s) and/or checking the fax, text, or email from the respondent.

All collected volumes are subjected to error checks during data validation. We verify data flagged for potential errors with the respondents. Volume data reported for only part of the year are inflated to approximate sales volumes for a full calendar year.

EIA-888, "On-Highway Diesel Fuel Price Survey"

After processing, the outlet prices are checked by a diesel validation program. The program identifies outliers and allows the analyst to further examine the data. Significant outliers are investigated and verified by calling the respondent(s) and/or checking the fax or email from the respondent for explanations. Also, credit card transaction prices are obtained from a private source and used to validate the U.S. and PADD level average prices for on-highway diesel fuel. If the survey results differ significantly from these sources, additional verification of the reported prices is done.

All collected volumes are subjected to error checks during data validation. We verify data flagged for potential errors with the respondents. Volume data reported for only part of the year are inflated to approximate sales volumes for a full calendar year.

Dissemination

Estimates of average retail gasoline and on-highway diesel prices are released at approximately 5:00 p.m. each Monday, except on Federal holidays, in which case the data are released on Tuesday (but still represent Monday's prices). These estimates are released on EIA's website: http://www.eia.gov/petroleum/gasdiesel/.

The data are also available through email notification to those customers who sign up for that service. The U.S., PADD, and sub-PADD level regular gasoline and diesel fuel average price estimates are available in this publication, the *Weekly Petroleum Status Report*.

Quality

Response Rates

For EIA-878 and EIA-888, EIA calculates the survey response rate based on the annual volumes represented by the reporting outlets in the sample. For EIA-878, in terms of total weighted annual sales volume, the volumes represented by the reporting outlets in the weekly survey account for at least 80% for regular grade at the U.S. level. For EIA-888, in terms of total weighted annual sales volume, the volumes represented by the reporting outlets in the weekly survey account for at least 80% of diesel fuel sold at the contiguous U.S. level.

Sampling and Non-sampling Errors

Sampling Errors

Sampling error is a statistical term for the error that occurs when survey estimates are based on a sample rather than being derived from a complete census of the frame. Tables showing data from the EIA-878 and EIA-888 surveys utilize a sample of resellers and retailers and, therefore, have sampling error.

Statisticians use measures of sampling variability, such as the standard Statisticians use measures of sampling variability, such as the standard error and the coefficient of variation, to measure the sampling error. These measures of sampling variability are typically estimated from the sample that was selected. The standard error, which is measured in the same units (current dollars per gallon for weekly gasoline and diesel prices) as the estimate, is a measure of the sampling variability of the estimate based on all possible samples that could have been selected using the chosen sample design. The coefficient of variation, which may also be referred to as the relative standard error, is the standard error expressed as a fraction of the estimate. For quality assurance purposes, we flag average price estimates if the corresponding estimated coefficient of variation is more than 5%.

Estimated measures of sampling variability for the EIA-878 can be found at: https://www.eia.gov/petroleum/gasdiesel/gas_proc-methods.php

Estimated measures of sampling variability for the EIA-888 can be found at: https://www.eia.gov/petroleum/gasdiesel/diesel_proc-methods.php

Non-sampling Errors

Potential errors unrelated to sampling, called nonsampling errors, include various response and operational errors, such as those related to data collection, respondent reporting, transcription, and nonresponse. All these types of errors could also occur even if every known outlet had been surveyed under the same conditions as the sample survey. Although nonsampling error is not measured directly, EIA employs quality control procedures throughout the survey process.

Revision Policy

EIA disseminates revised weekly data only if the revision is expected to

substantively affect users understanding of U.S. petroleum prices. The decision to disseminate a revision to weekly data will be based on EIA's judgment of the revision's expected effect. If a revision is necessary, it will be disseminated in the next regularly scheduled release of the weekly products.

Confidentiality—Data protection and disclosure for Weekly Price Surveys

The information reported on the weekly price survey Forms EIA-878 and EIA-888 is considered confidential in accordance with the Confidential Information Protection and Statistical Efficiency Act of 2018 (P.L. 107-347) and the information will be used solely for statistical purposes. Instructions to the forms include the following:

"The information you provide will be used for statistical purposes only and is confidential by law. In accordance with the Confidential Information Protection and Statistical Efficiency Act of 2018 and other applicable Federal laws, your responses will not be disclosed in identifiable form without your consent. Per the Federal Cybersecurity Enhancement Act of 2015, Federal information systems are protected from malicious activities through cybersecurity screening of transmitted data. Every EIA employee, as well as every agent, is subject to a jail term, a fine, or both if he or she makes public ANY identifiable information you reported."

Notes

Note 1

Calculation of World Oil Price

The weighted average international price of oil, shown in the Highlights and in Table 10, is an average calculated using specific crude oil prices weighted by the estimated crude oil export volume for each oil-producing country. To develop Table 10, a list of major oil producing/ exporting countries was chosen. For each country, the contract selling price of one or more representative crude oils was determined by investigating a number of industry publications (i.e., Platt's Oilgram Price Report, Wall Street Journal, and Canadian Ministry of Natural Resources) and by contacting oil market analysts. Then, the appropriate crude oil exporting volumes to be used as weighting factors for each country were determined. These volumes are estimates based on a number of sources which provide data on production, consumption, and petroleum product exports for these countries. Export volumes for a number of smaller producing/ exporting countries, not listed in the table, are included in the weighting factors. After the export volumes had been determined, simple mathematical weighted averages were calculated to arrive at the Total OPEC, Total Non-OPEC, and Total World prices. The average United States (FOB) import price is derived by the same basic procedure as the world oil price that is, taking the representative contract crude oil price of a specific crude oil from a particular country and weighting this price by a certain volume of crude oil. In this case, the weighting factors are the volumes of crude oil imported into the U.S. from pertinent countries. Import volumes from a number of smaller producing/exporting countries, not listed in the table, are included in the weighting factors.

Note 2

The spot prices that are shown in Tables 11 and 12 are calculated by taking an unweighted average of the daily closing spot prices for a given product over a specified time period, such as a week or month.

Note 3

The futures prices shown in Table 13 are the official daily closing prices at 2:30 p.m. from the trading floor of the New York Mercantile Exchange (NYMEX) for a specific delivery month for each product listed.

Note 4

The futures price differentials shown in Figure 9 show the market premium for the first NYMEX delivery month contract over the second. For example, the data for September show the difference between October and November futures contract prices for crude oil and petroleum products, indicating the relative values placed by markets on commodities to be delivered during those two months. This differential, if negative and large enough, provides incentive for refiners and traders to hold product in storage, and if positive, to defer purchases until some future point in time.

Note 5

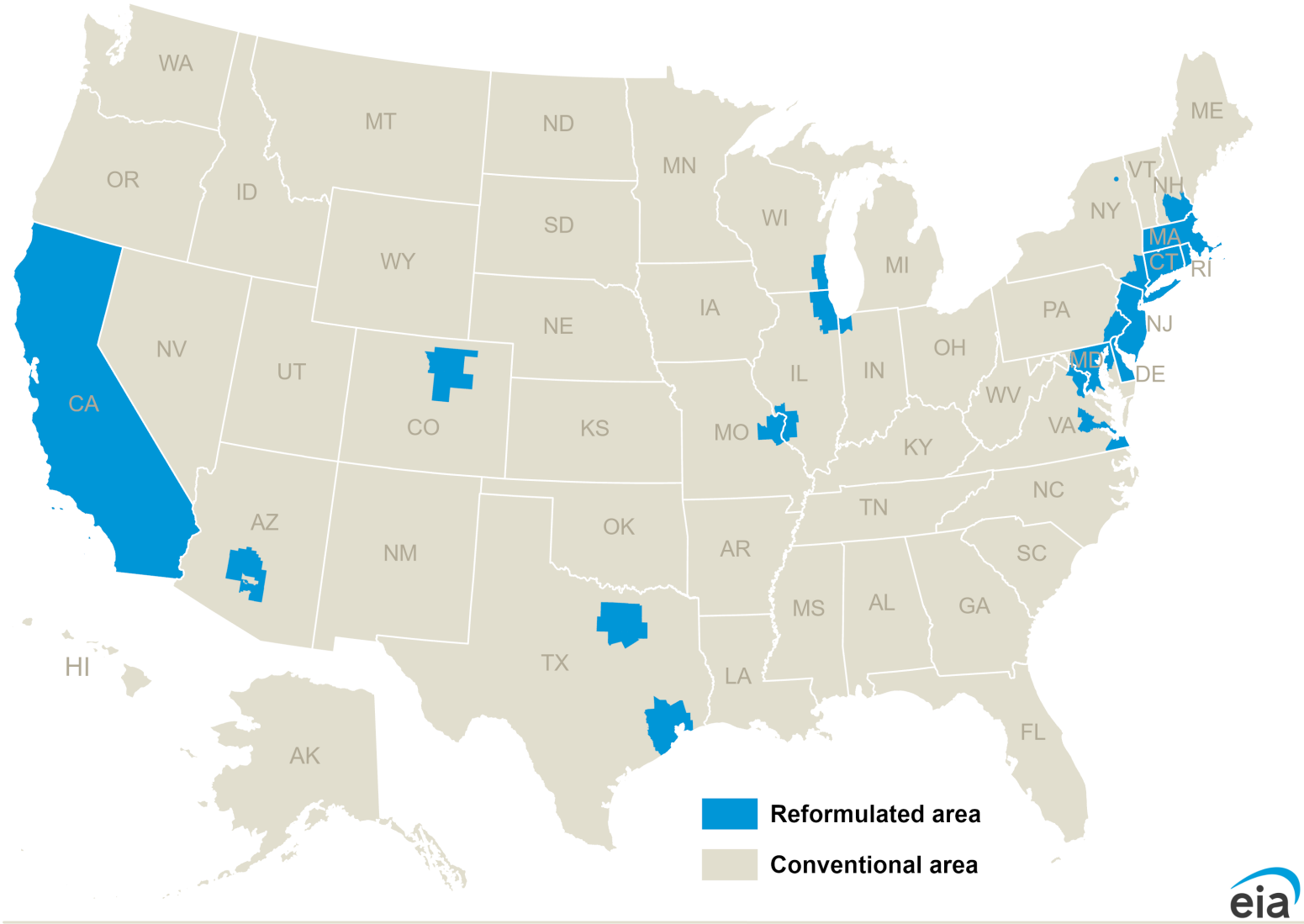
The retail gasoline prices shown in Table 14 reflect sales of reformulated gasoline (RFG) in those areas where required by Federal or State law and conventional gasoline elsewhere (see Figure B1). Areas requiring RFG may change over time due to the ozone non¬attainment status of an area being re-designated by the Environmental Protection Agency (EPA), a State opting in or out of an EPA clean fuel program, or a State adopting its own specific clean fuel program. EIA reclassifies the outlets reporting retail gasoline prices each time an area shifts in or out of a reformulated gasoline program. Conventional areas include areas where oxygenated gasoline may be required for all or part of the year.

Note 6

As a result of a processing error, we underestimated export quantities for most products (including crude oil, finished motor gasoline, distillate fuel oil, propane and propylene, and other oils) in WPSR for November 2021 through February 2022. Data users should exercise caution when using these export and product supplied estimates from WPSR as the basis for annual comparisons in their analyses. However, we believe that the corresponding PSM export and product supplied estimates for November 2021 through February 2022, which reflect official U.S. export statistics published by the U.S. Census Bureau, are accurate and should be used as the basis for annual comparisons involving export and product supplied data.

Figure B1. Gasoline Formulation Required by Area as of May 2026

Reformulated gasoline areas



Data source: U.S. Environmental Protection Agency and State environmental offices.